Azure Data Factory Capstone – Covid Use Case

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COACH – LAVANYA PALANI

PROBLEM STATEMENT

- How to ingest data from flat files into Azure Data Lake Gen2 and Azure Synapse using Azure Data Factory (ADF)
- How to transform data using Data Flows in Azure Data Factory (ADF) and load into Azure Synapse

INTRODUCTION

CREATING RESOURCE GROUP

CREATING AZURE DATA LAKE STORAGE GEN2 ACCOUNT

CREATING AZURE DATA LAKE STORAGE GEN2 CONTAINER

CREATING AZURE DATA FACTORY

CREATING AZURE SYNAPSE WORKSPACE

REQUIREMENTS

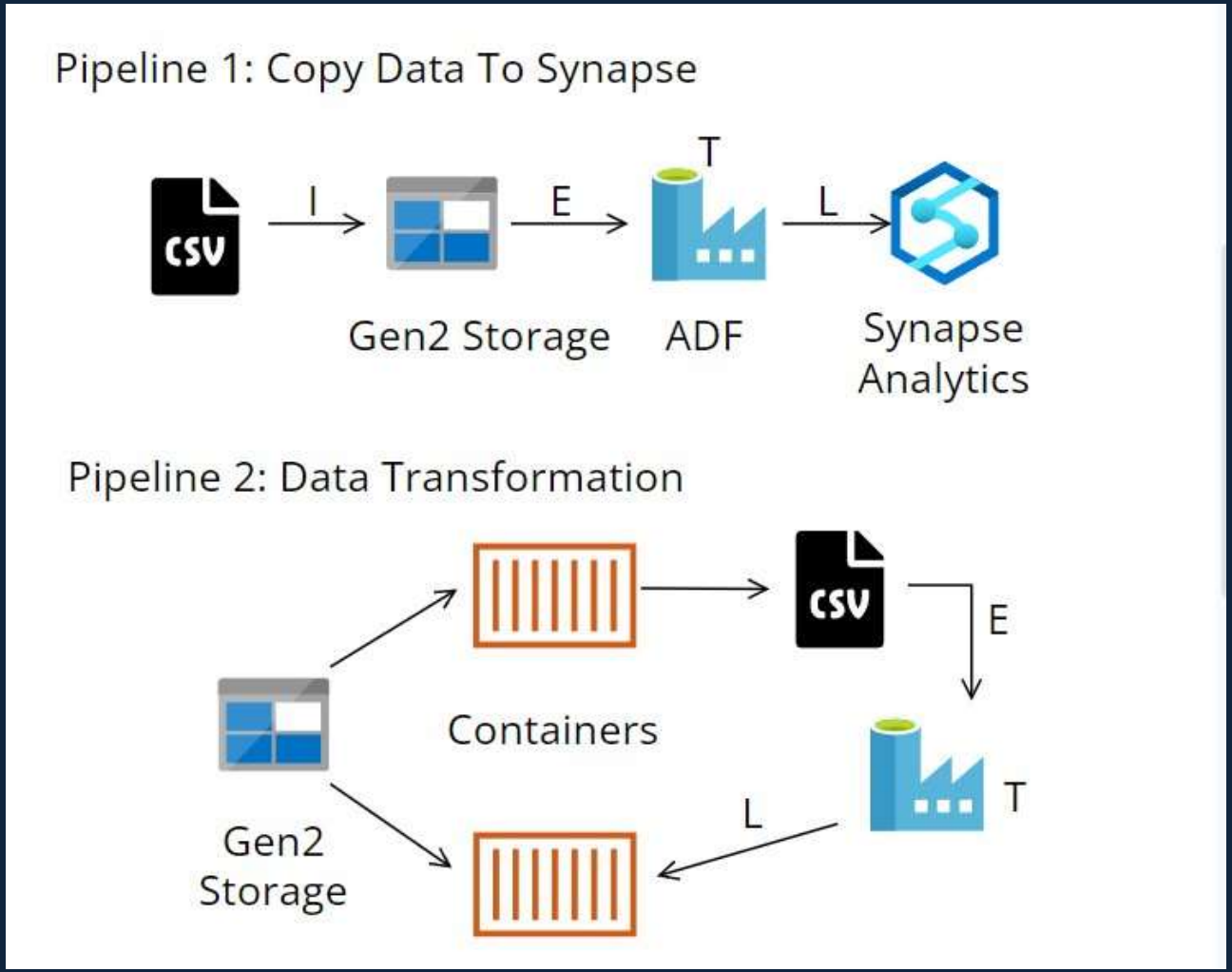
RESOURCE GROUPS

STORAGE ACCOUNTS

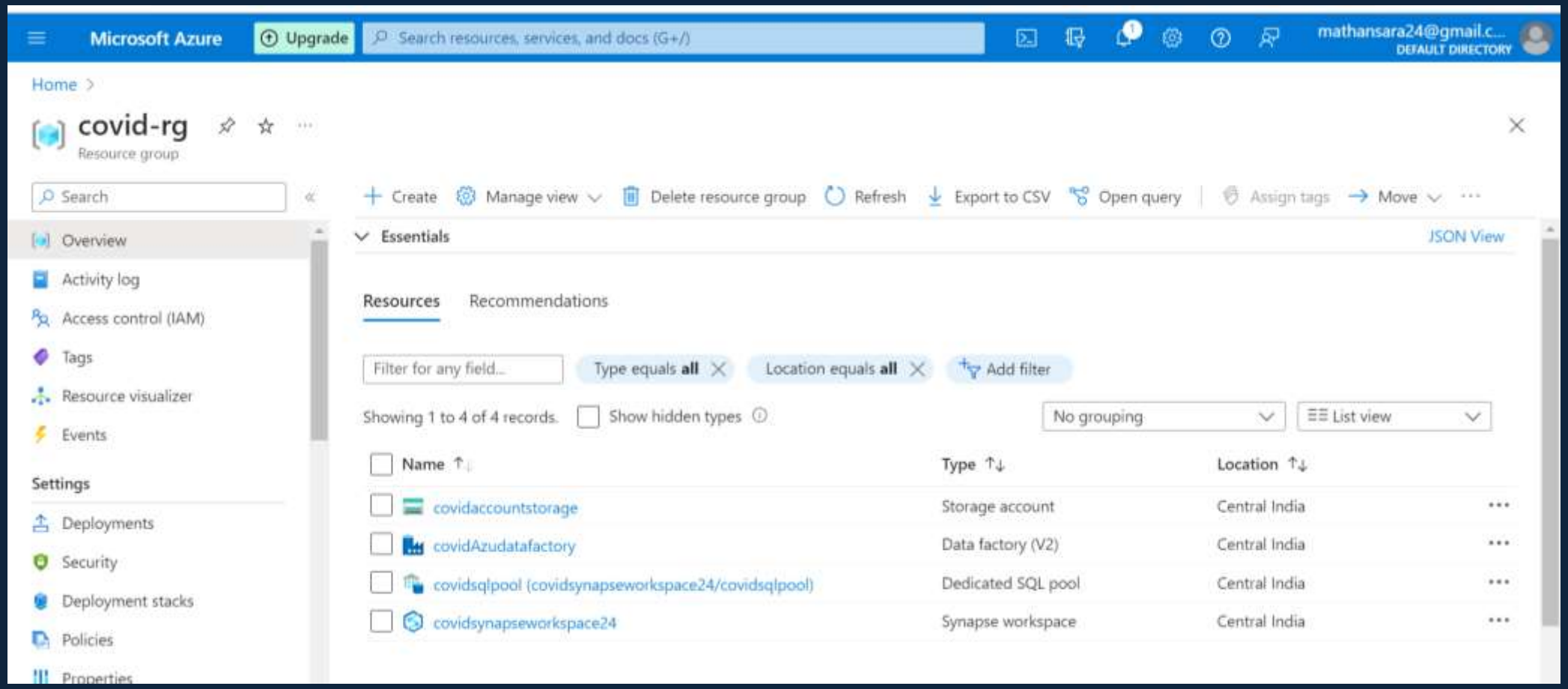
AZURE DATA FACTORY

AZURE SYNAPSE STUDIO

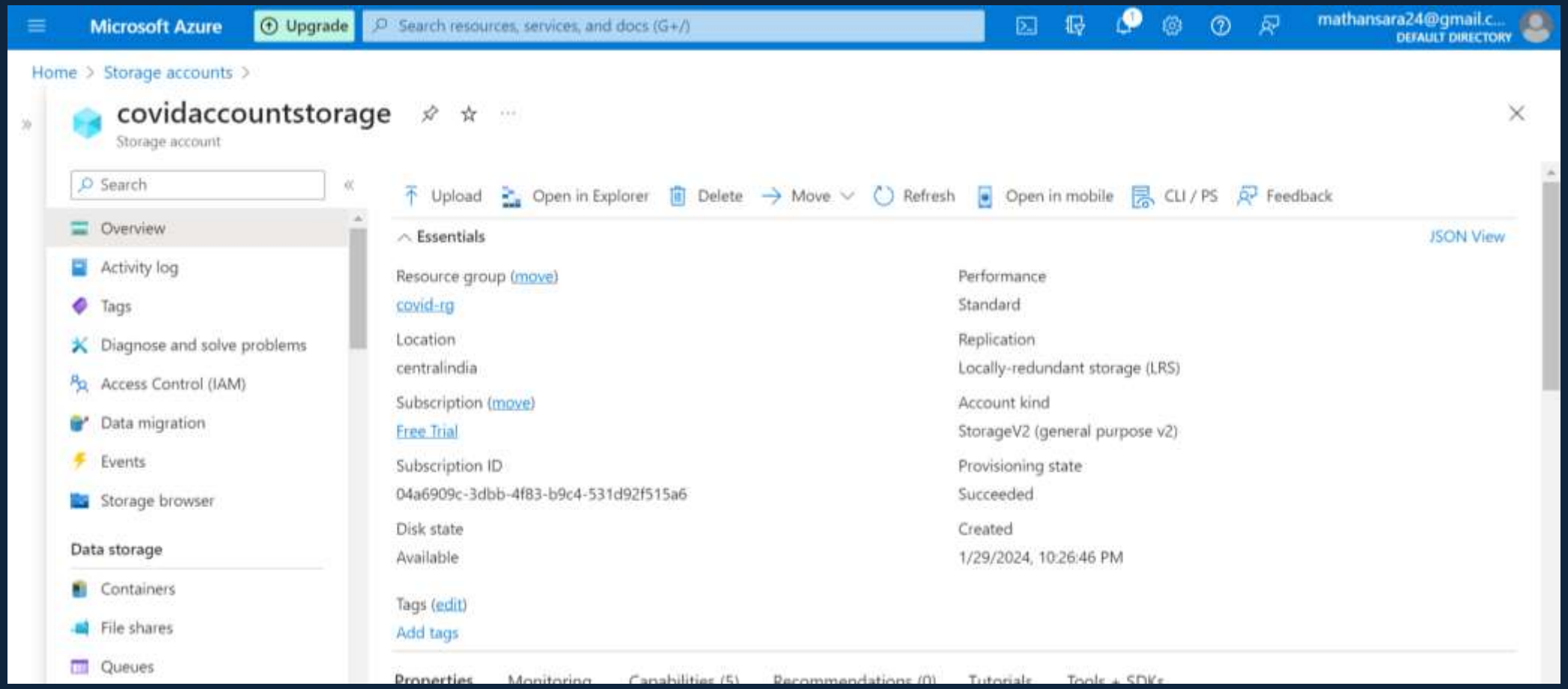
ARCHITECTURE DIAGRAM



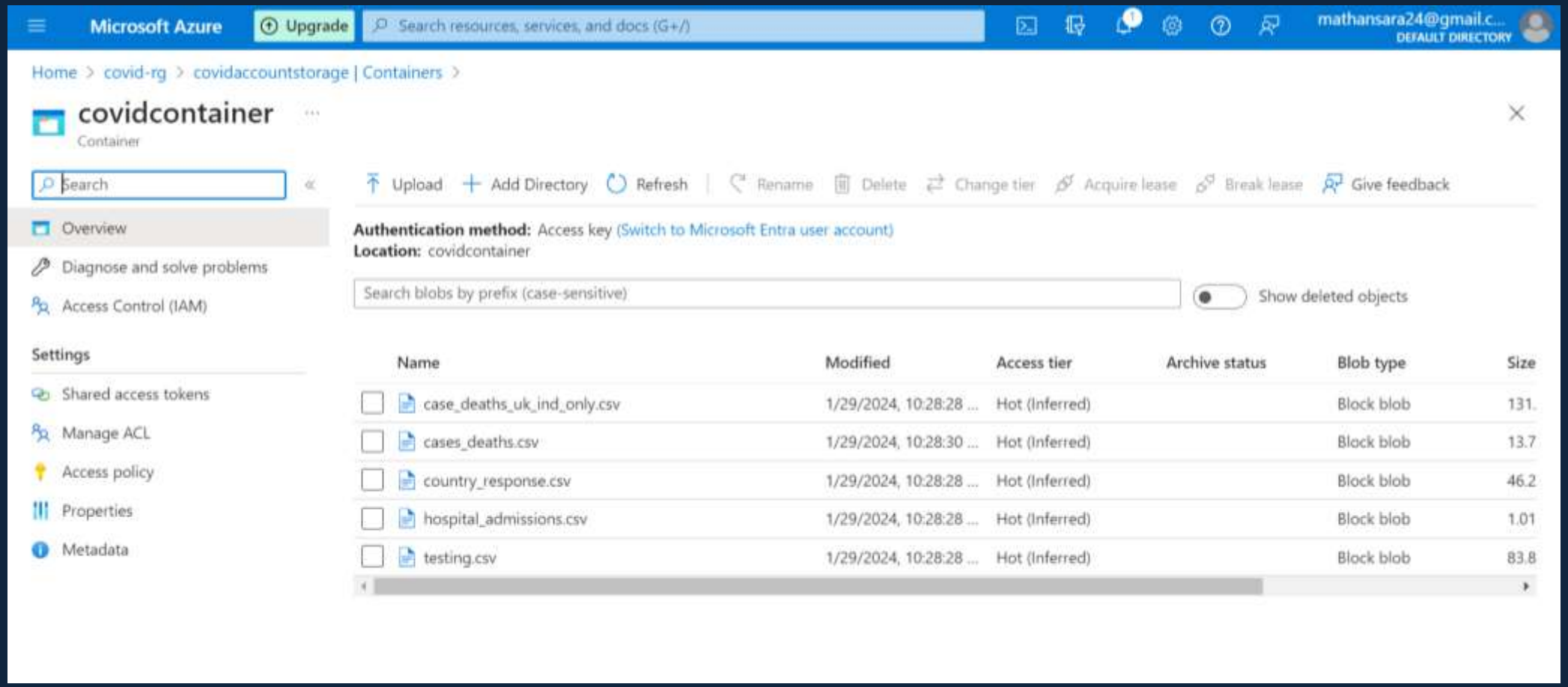
RESOURCE GROUP



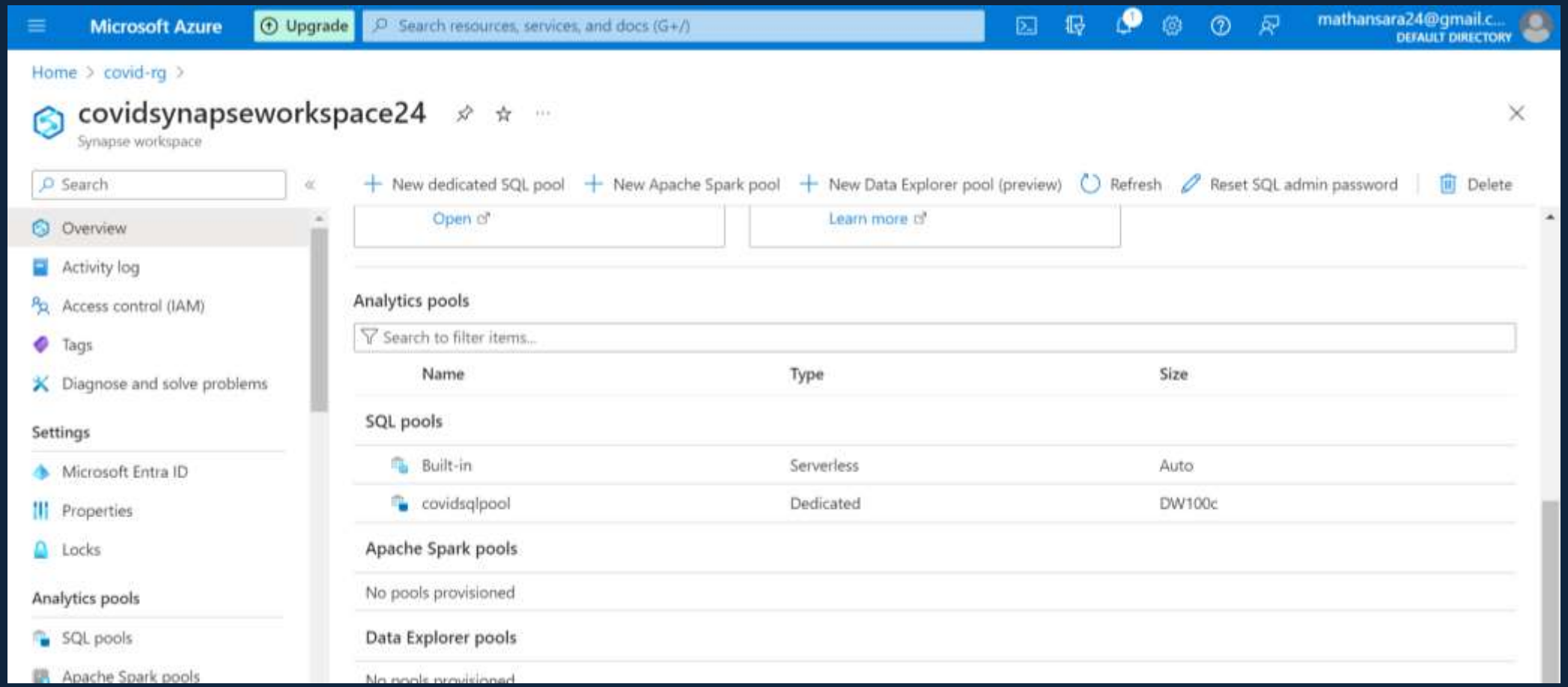
AZURE DATA LAKE STORAGE GEN2



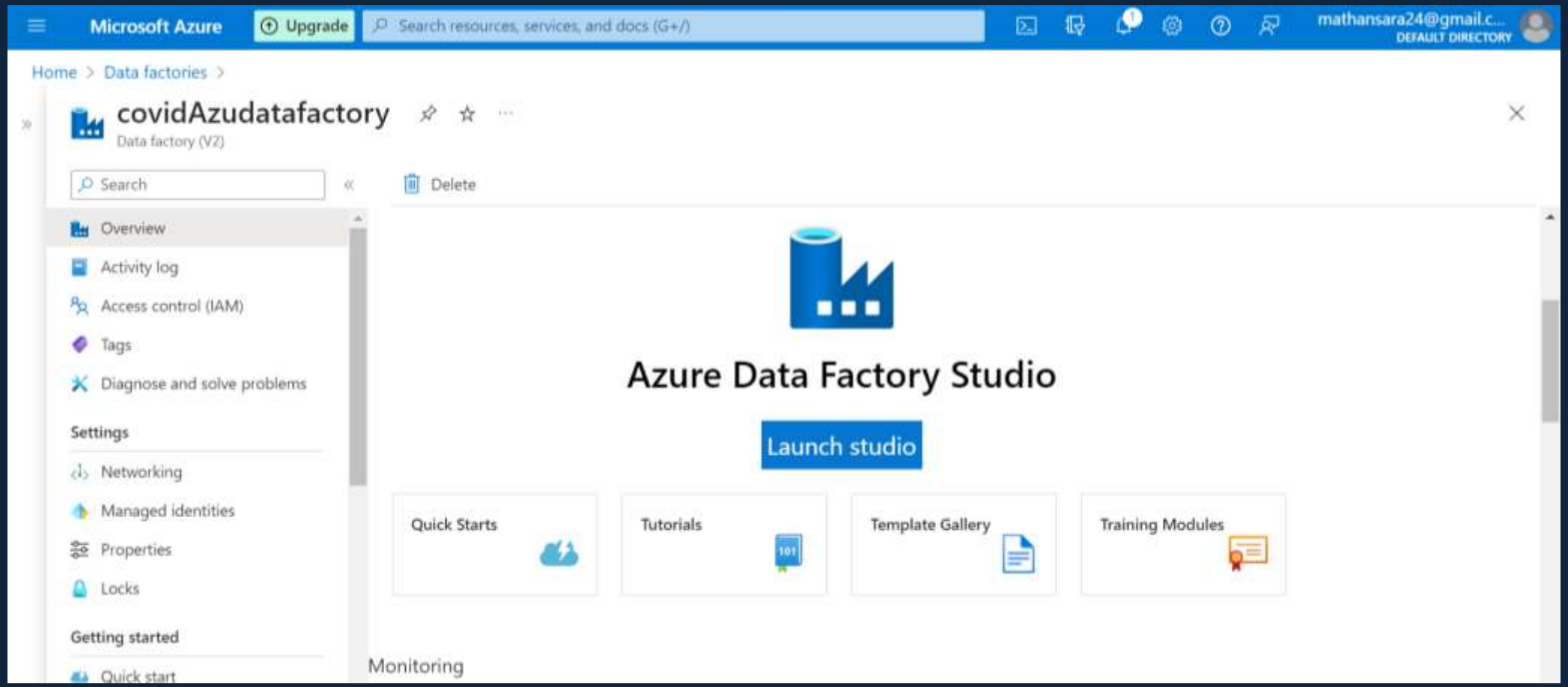
AZURE CONTAINER



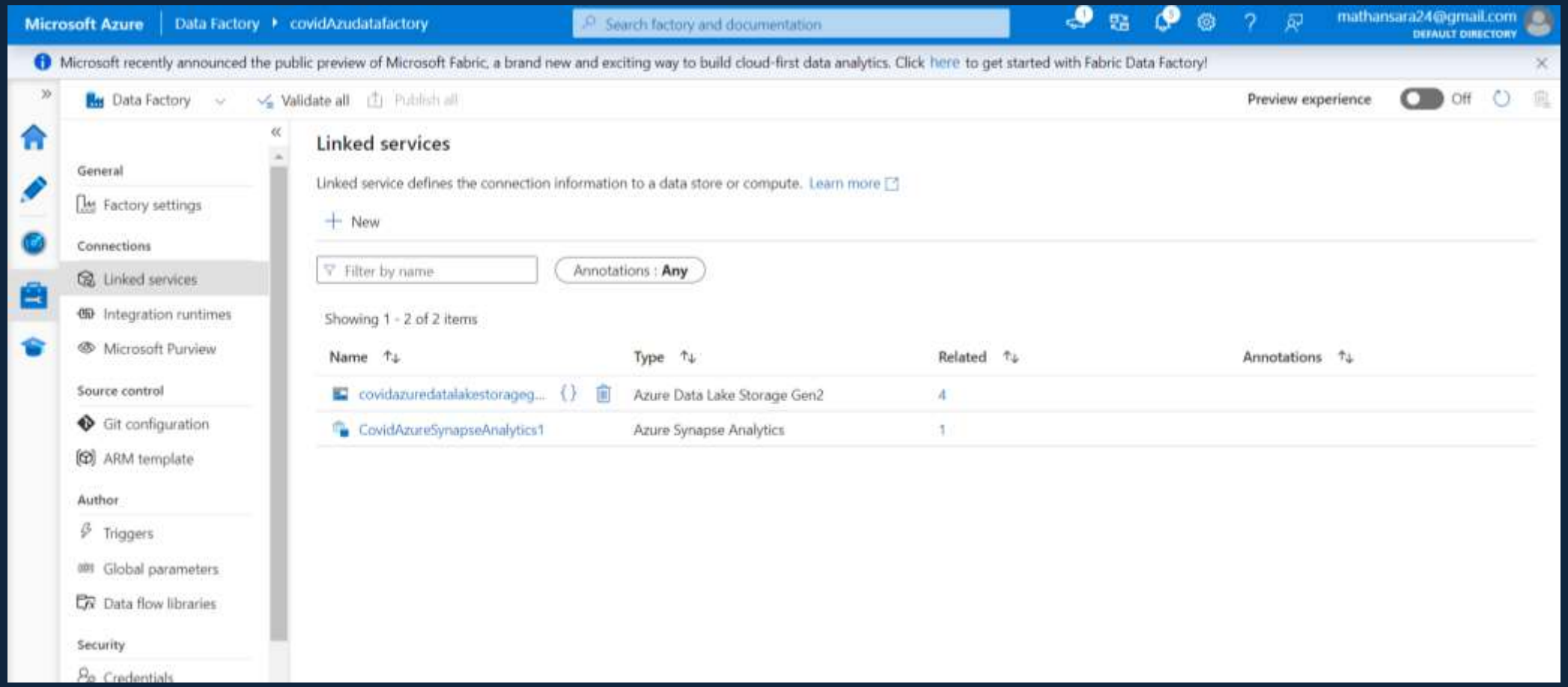
AZURE SYNAPSE WORKSPACE



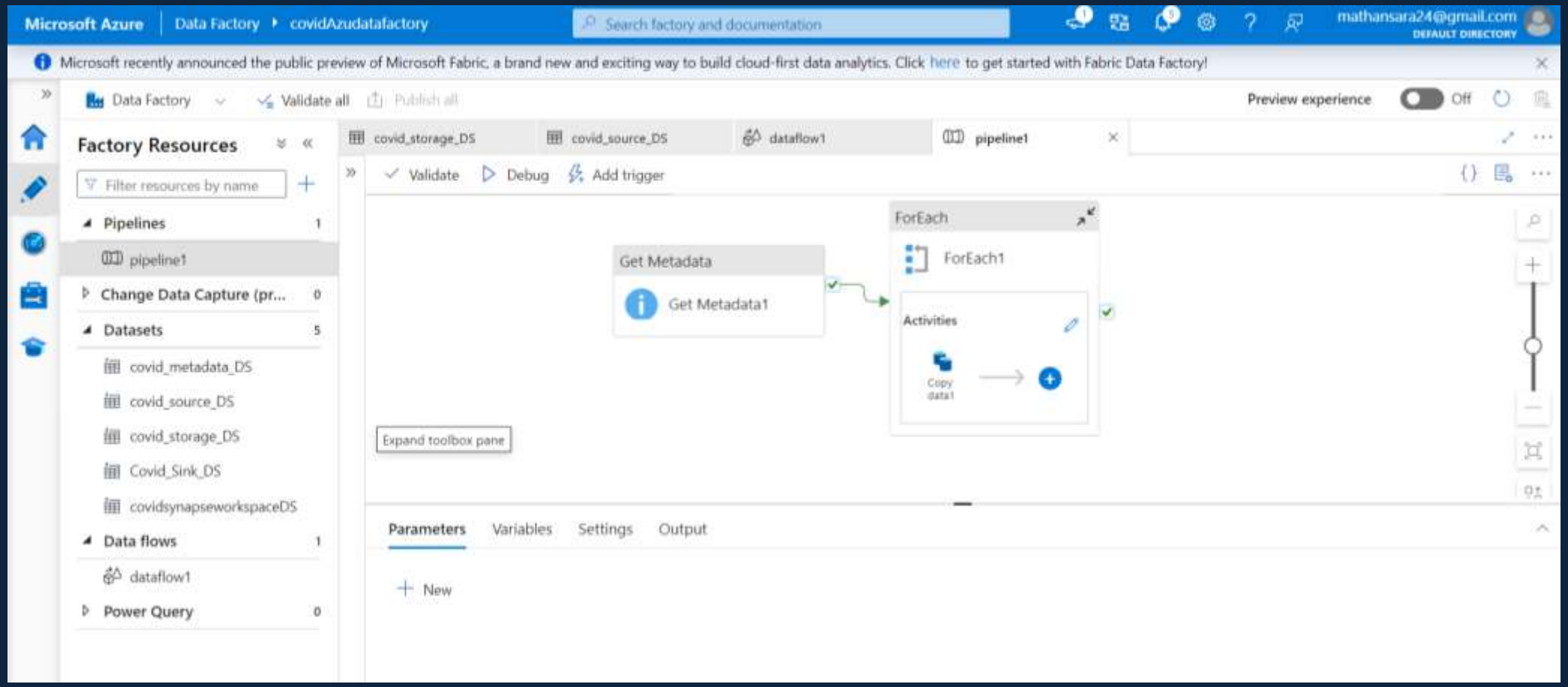
AZURE DATA FACTORY



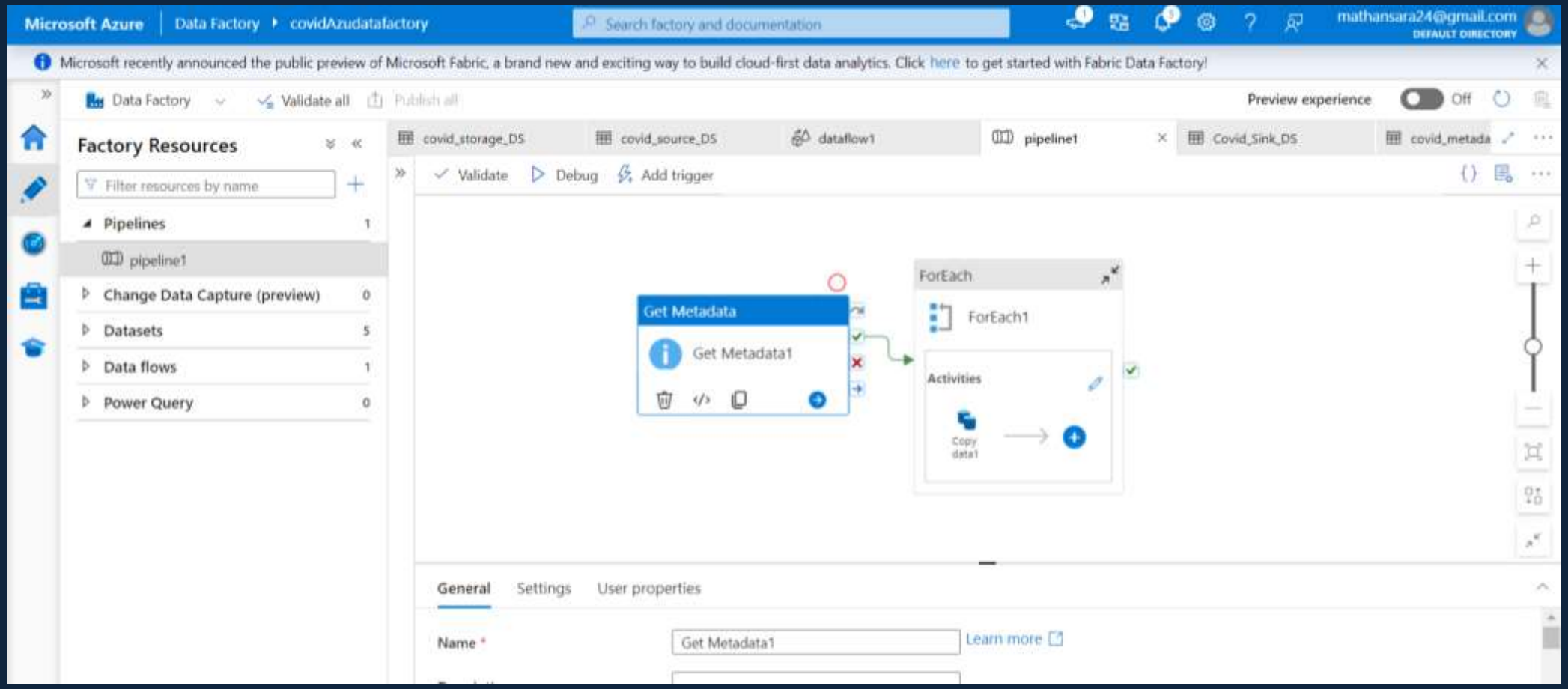
LINKED SERVICE FOR SOURCE AND SINK IN ADF



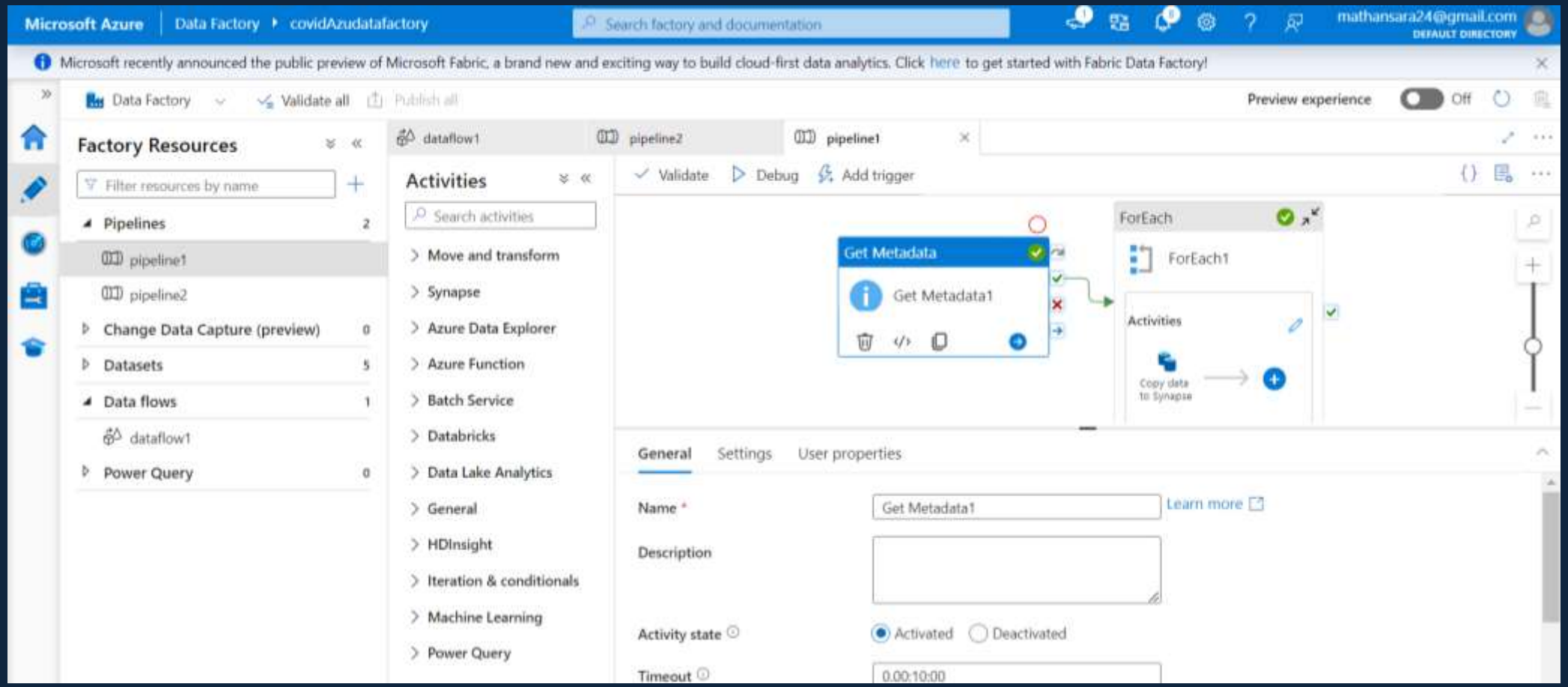
CREATED DATASETS



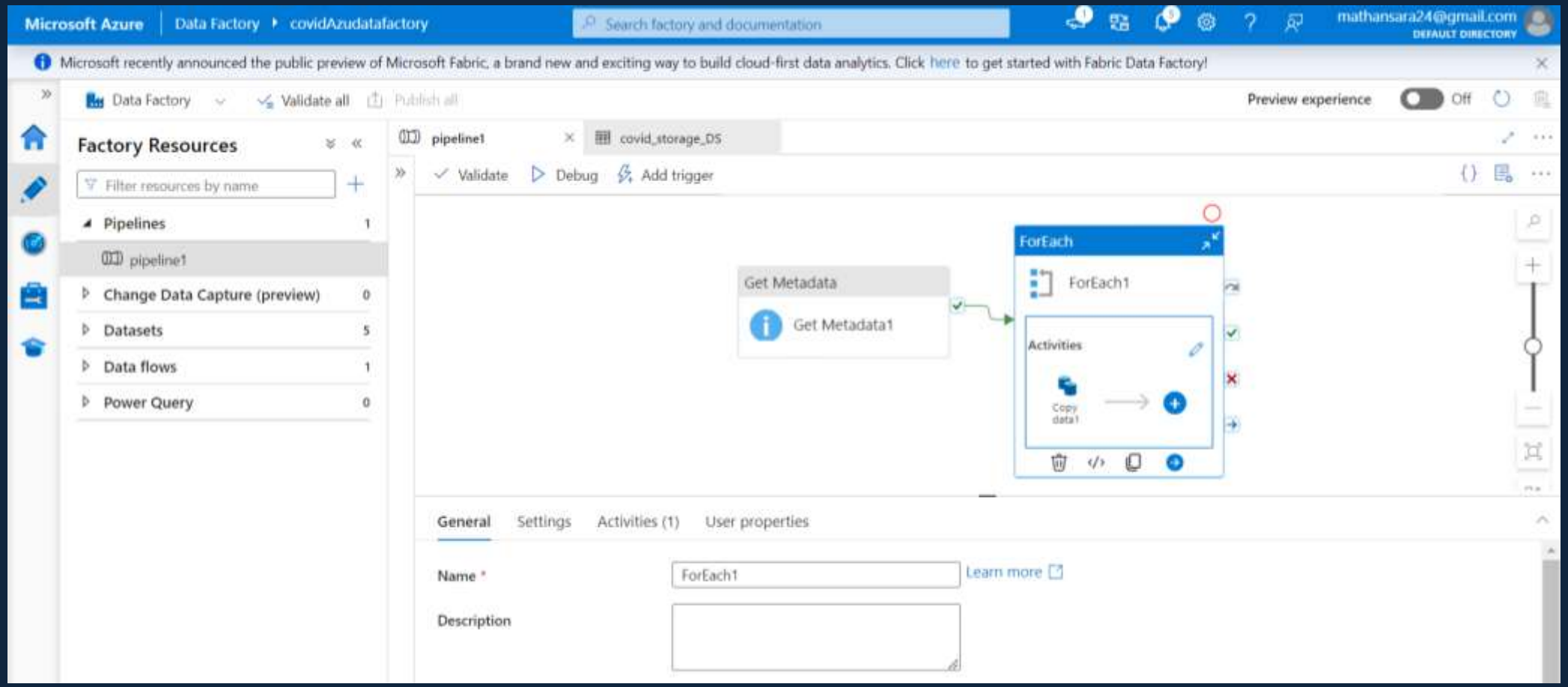
CREATED PIPELINE



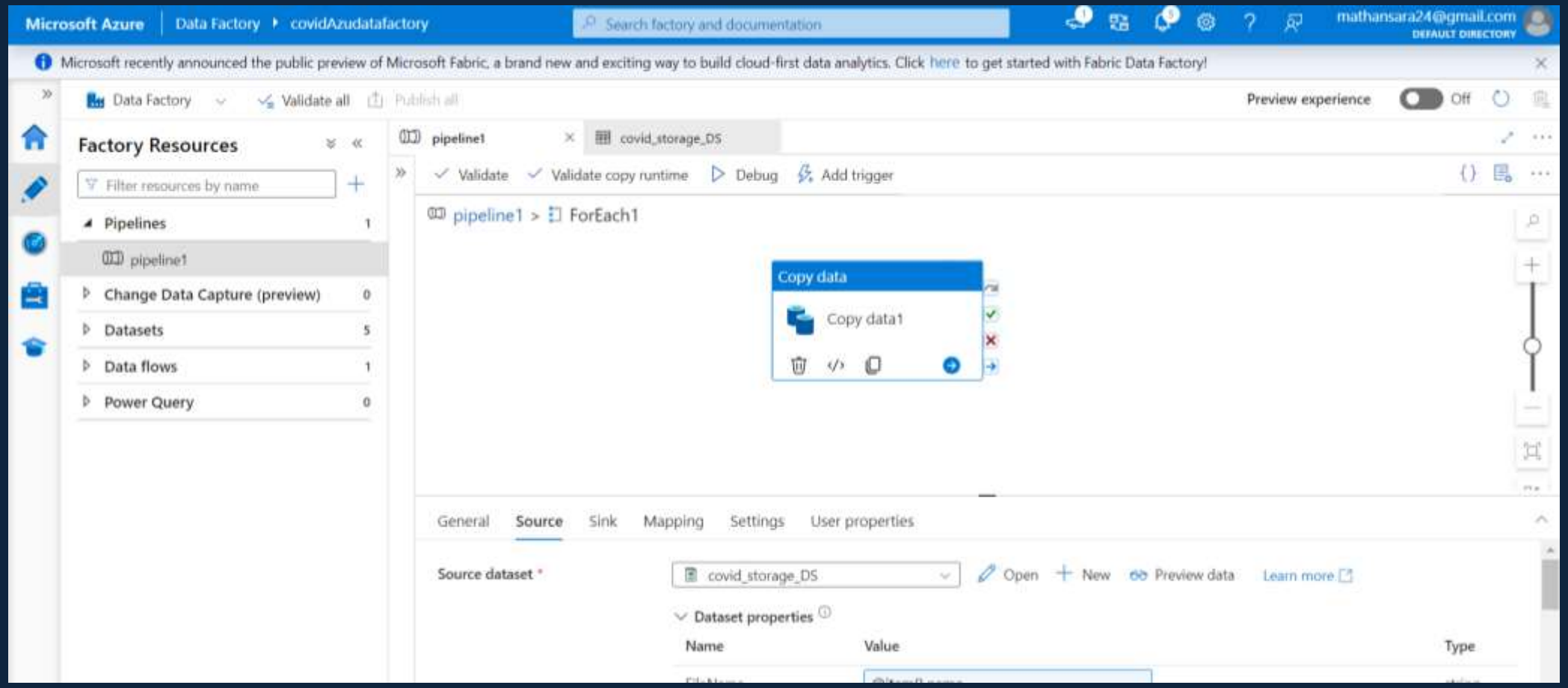
GET METADATA ACTIVITY



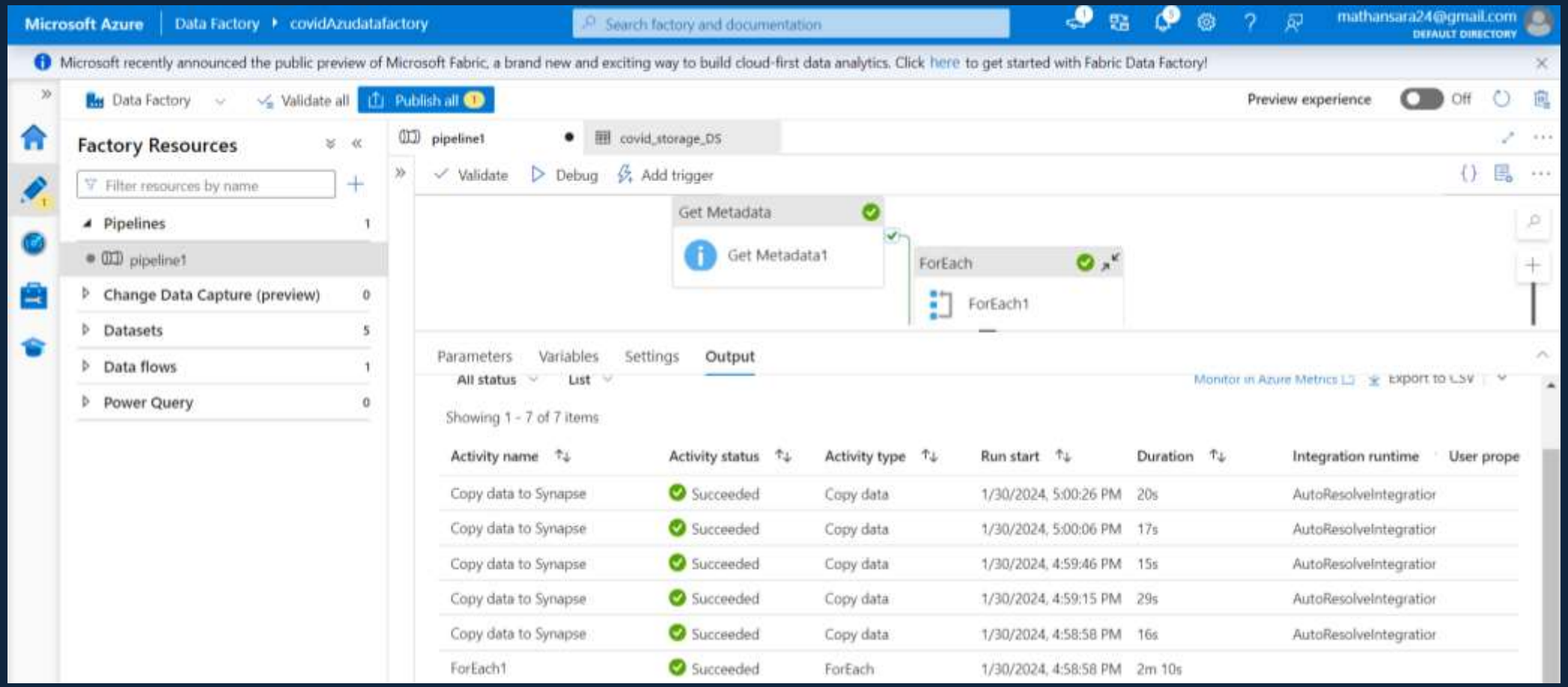
FOR EACH ACTIVITY



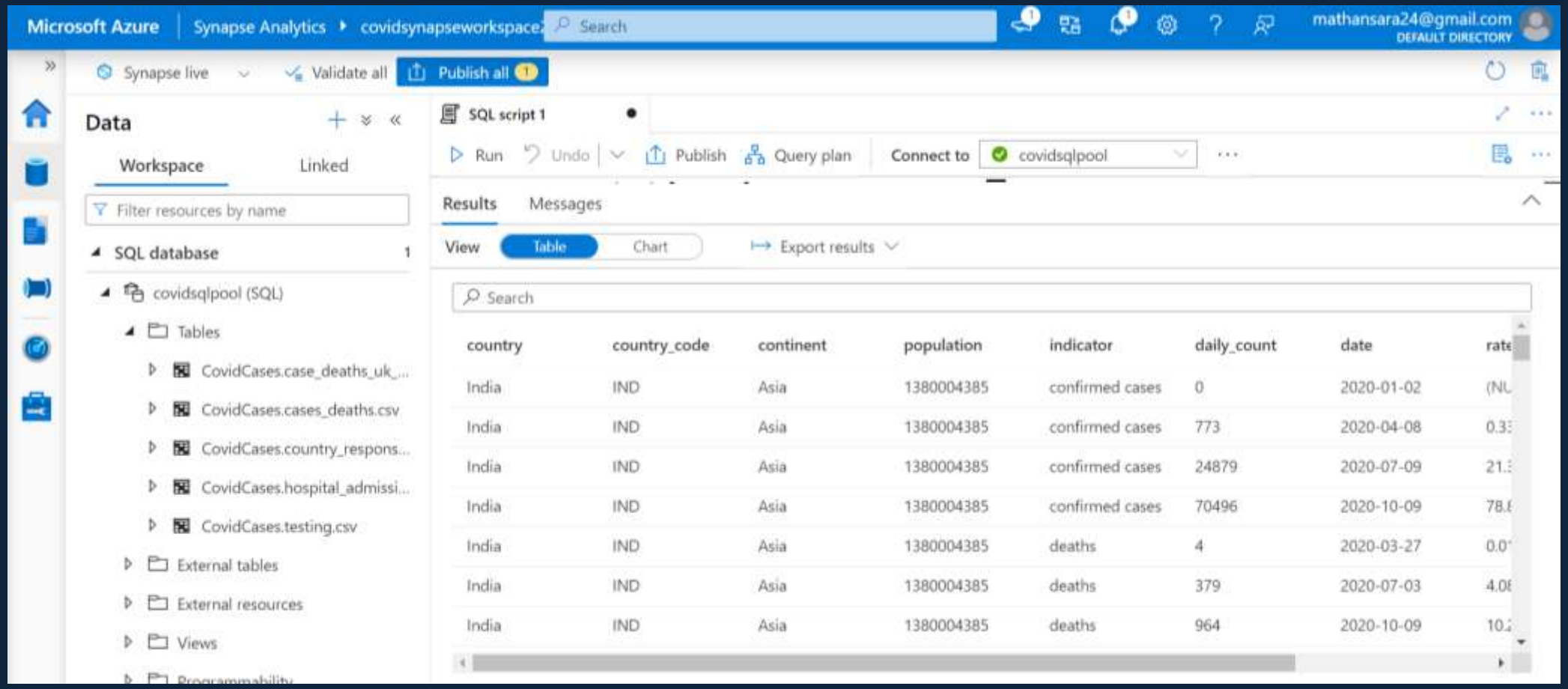
COPY DATA ACTIVITY



PIPELINE RESULTS



COPIED DATA TO SYNAPSE



country	country_code	continent	population	indicator	daily_count	date	rate
India	IND	Asia	1380004385	confirmed cases	0	2020-01-02	(NU
India	IND	Asia	1380004385	confirmed cases	773	2020-04-08	0.3
India	IND	Asia	1380004385	confirmed cases	24879	2020-07-09	21.
India	IND	Asia	1380004385	confirmed cases	70496	2020-10-09	78.
India	IND	Asia	1380004385	deaths	4	2020-03-27	0.0
India	IND	Asia	1380004385	deaths	379	2020-07-03	4.0
India	IND	Asia	1380004385	deaths	964	2020-10-09	10.

SCHEMA OF THE TABLE'S

Dbo.DeathsInUKAndIndia

Column Technical Name	Column Datatype	Nullable?	PK?	FK?	Column Notes
Country	Varchar	Yes			
CountryCode	Varchar	Yes			
Continent	Varchar	Yes			
Population	BigInt	Yes			
Indicator	Varchar	Yes			
DailyCount	BIgInt	Yes			
Date	Date	Yes			
Rate_14_Day	Decimal	Yes			
Source	Varchar	Yes			

Dbo. AllOverDeaths

Column Technical Name	Column Datatype	Nullable?	PK?	FK?	Column Notes
Country	Varchar	Yes			
CountryCode	Varchar	Yes			
Continent	Varchar	Yes			
Population	BigInt	Yes			
Indicator	Varchar	Yes			
DailyCount	BIgInt	Yes			
Date	Date	Yes			
Rate_14_Day	Decimal	Yes			
Source	Varchar	Yes			

Dbo.CountryWiseResponseMeasure

Column Technical Name	Column Datatype	Nullable?	PK?	FK?	Column Notes
Country	Varchar	Yes			
ResponseMeasure	Varchar	Yes			
DateStart	Varchar	Yes			
DateEnd	Varchar	Yes			

SCHEMA OF THE TABLE'S

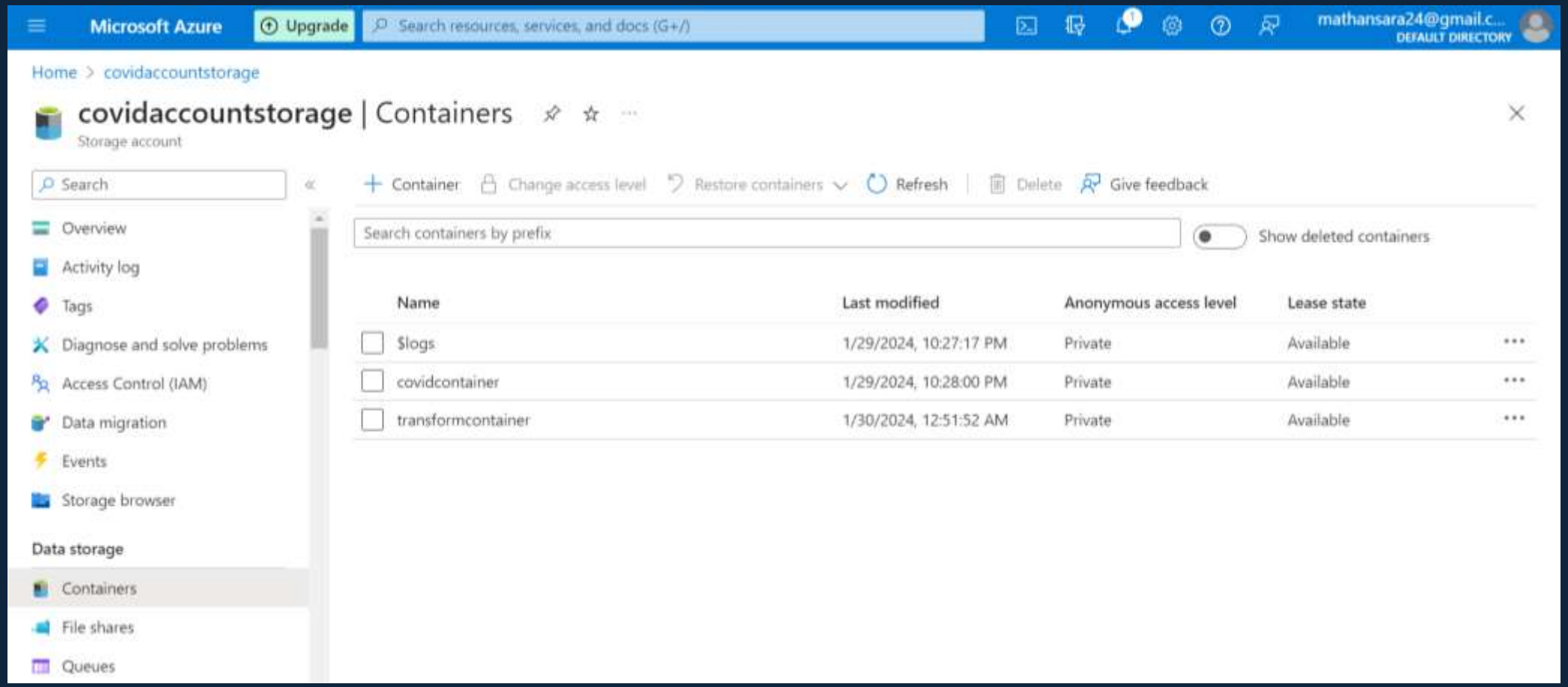
Dbo.HospitalAdmissions

Column Technical Name	Column Datatype	Nullable?	PK?	FK?	Column Notes
Country	Varchar	Yes			
Indicator	Varchar	Yes			
Date	Datetime	Yes			
Year_Week	Varchar	Yes			
Value	Decimal	Yes			
Source	Varchar	Yes			
URL	Varchar	Yes			

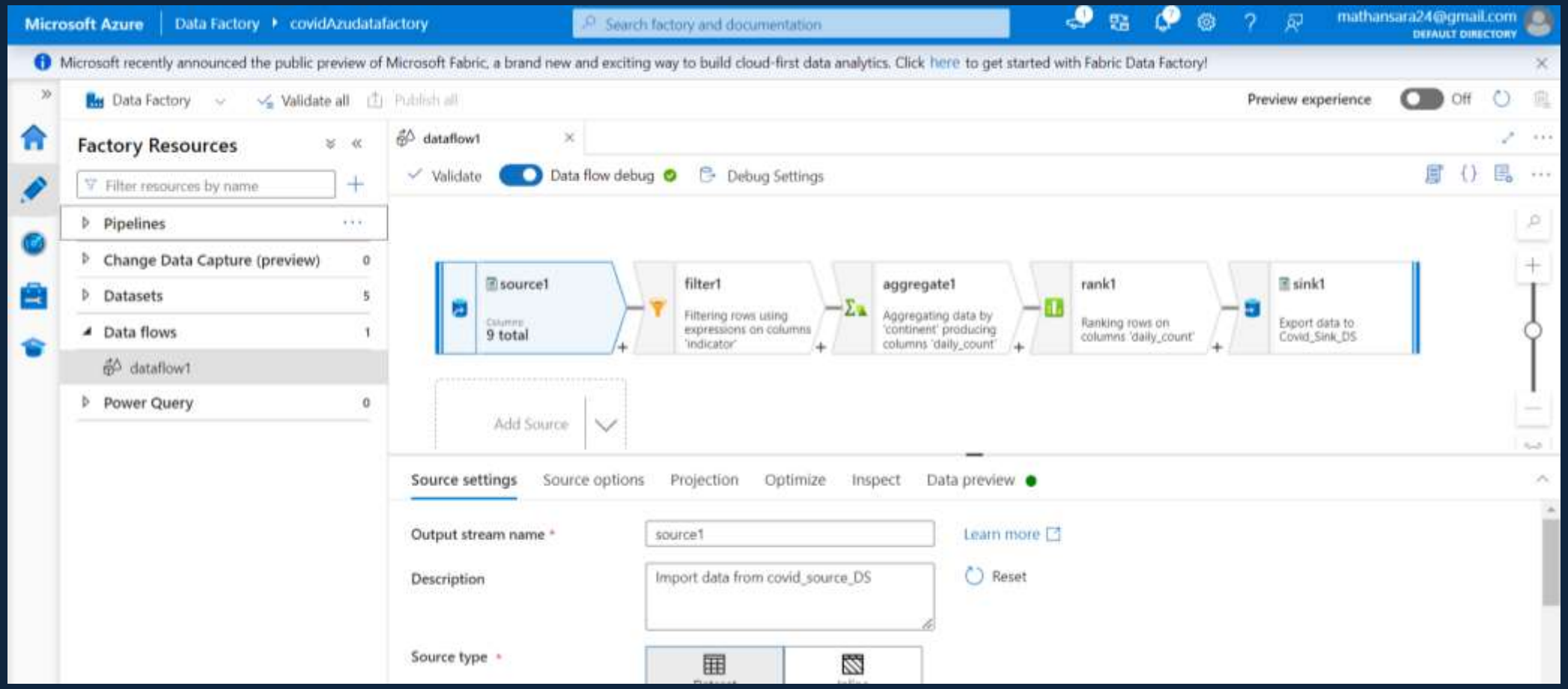
Dbo.Testing

Column Technical Name	Column Datatype	Nullable?	PK?	FK?	Column Notes
Country	Varchar	Yes			
CountryCode	Varchar	Yes			
Year_Week	Varchar	Yes			
New_Cases	BigInt	Yes			
Tests_Done	BigInt	Yes			
Population	BigInt	Yes			
Testing_Rate	Decimal	Yes			
Positivity_Rate	Decimal	Yes			
Testing_Data_Source	Varchar	Yes			

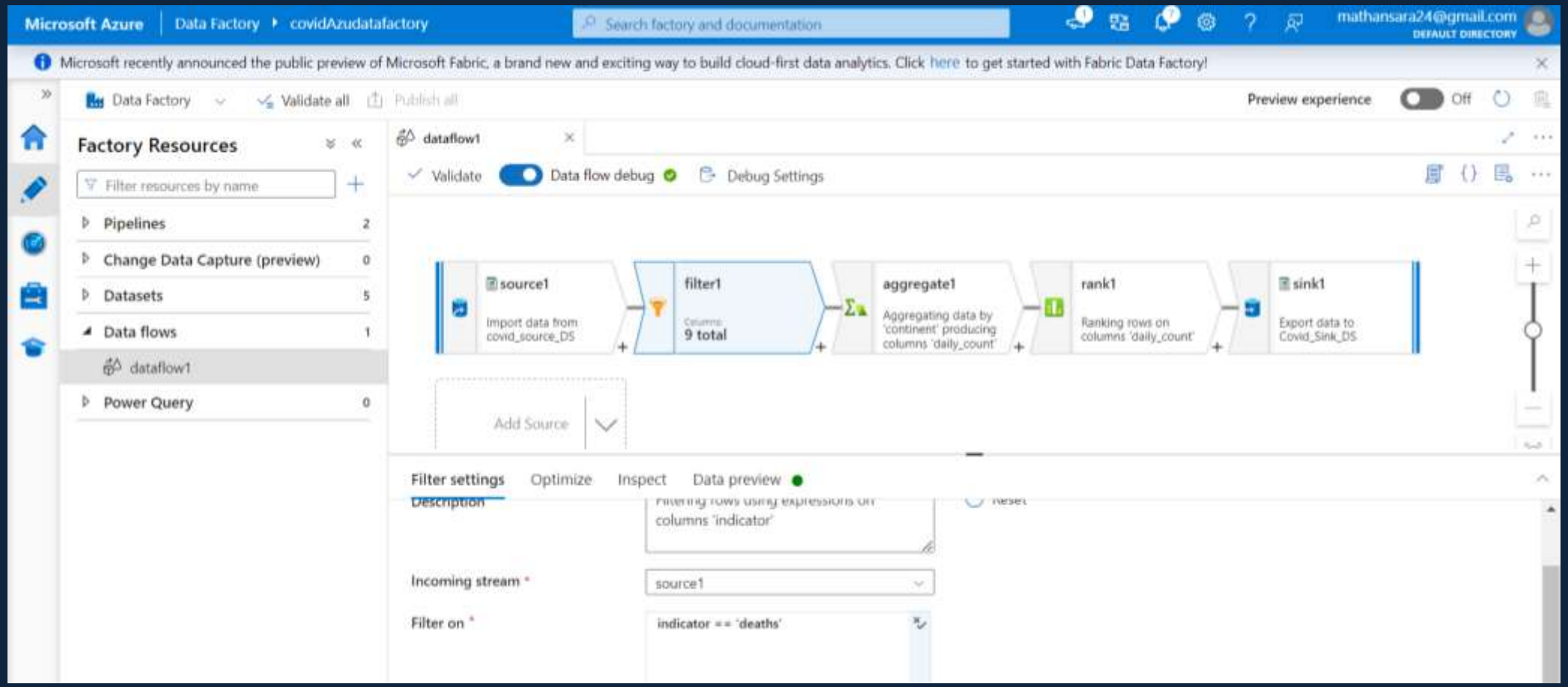
CREATED ANOTHER CONTAINER FOR STORING TRANSFORMED DATA IN GEN2



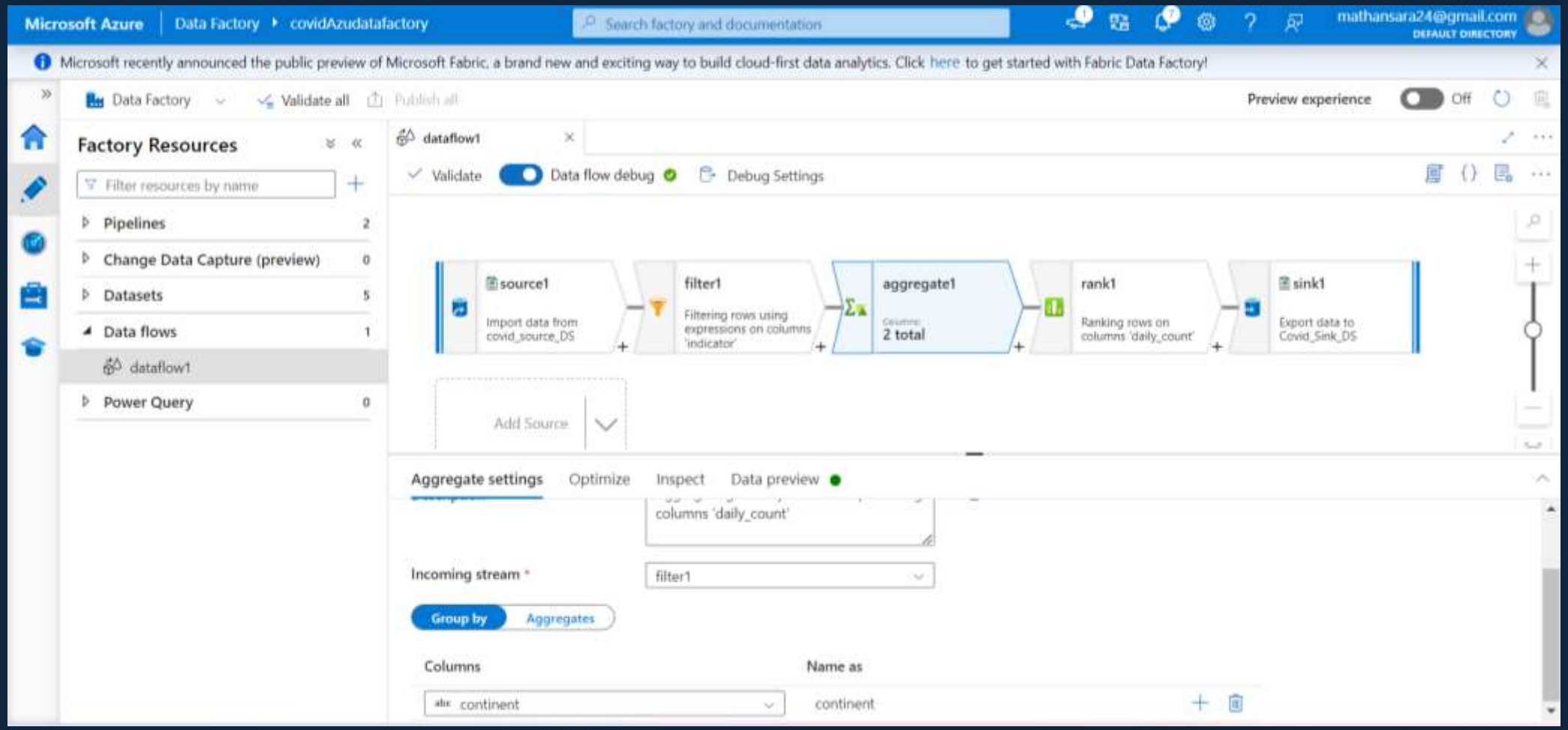
DATA FLOW – ADDING SOURCE



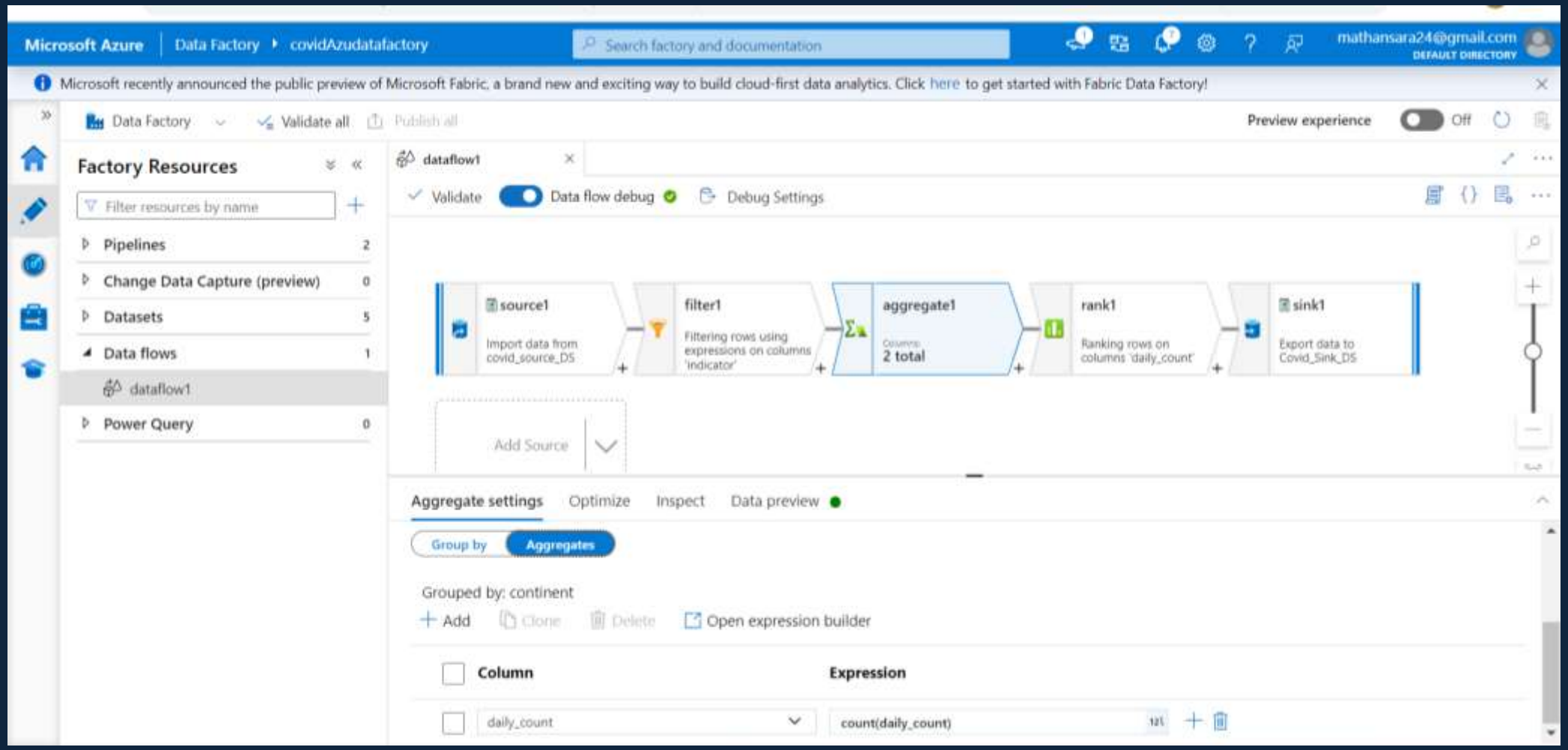
DATA FLOW – FILTER ACTIVITY



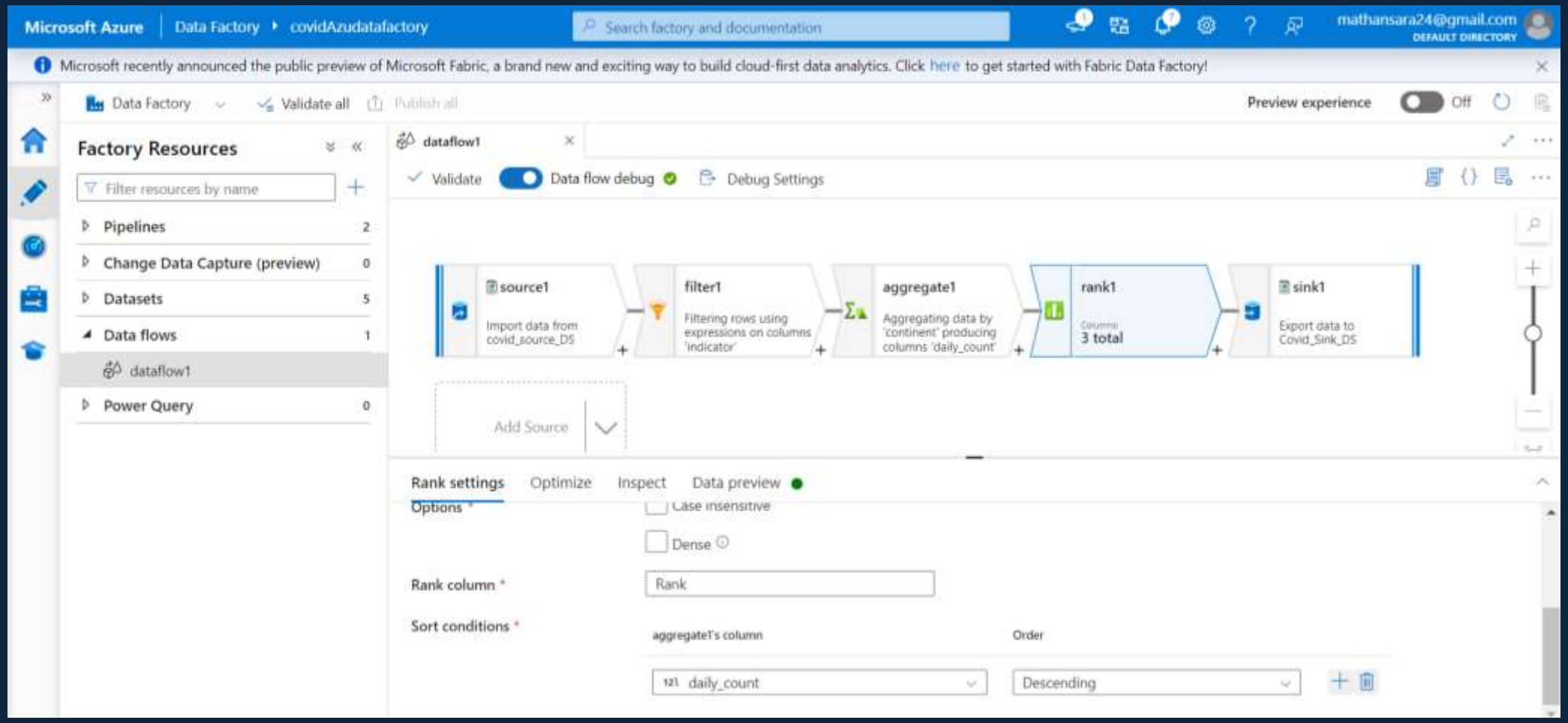
DATA FLOW – AGGREGATE



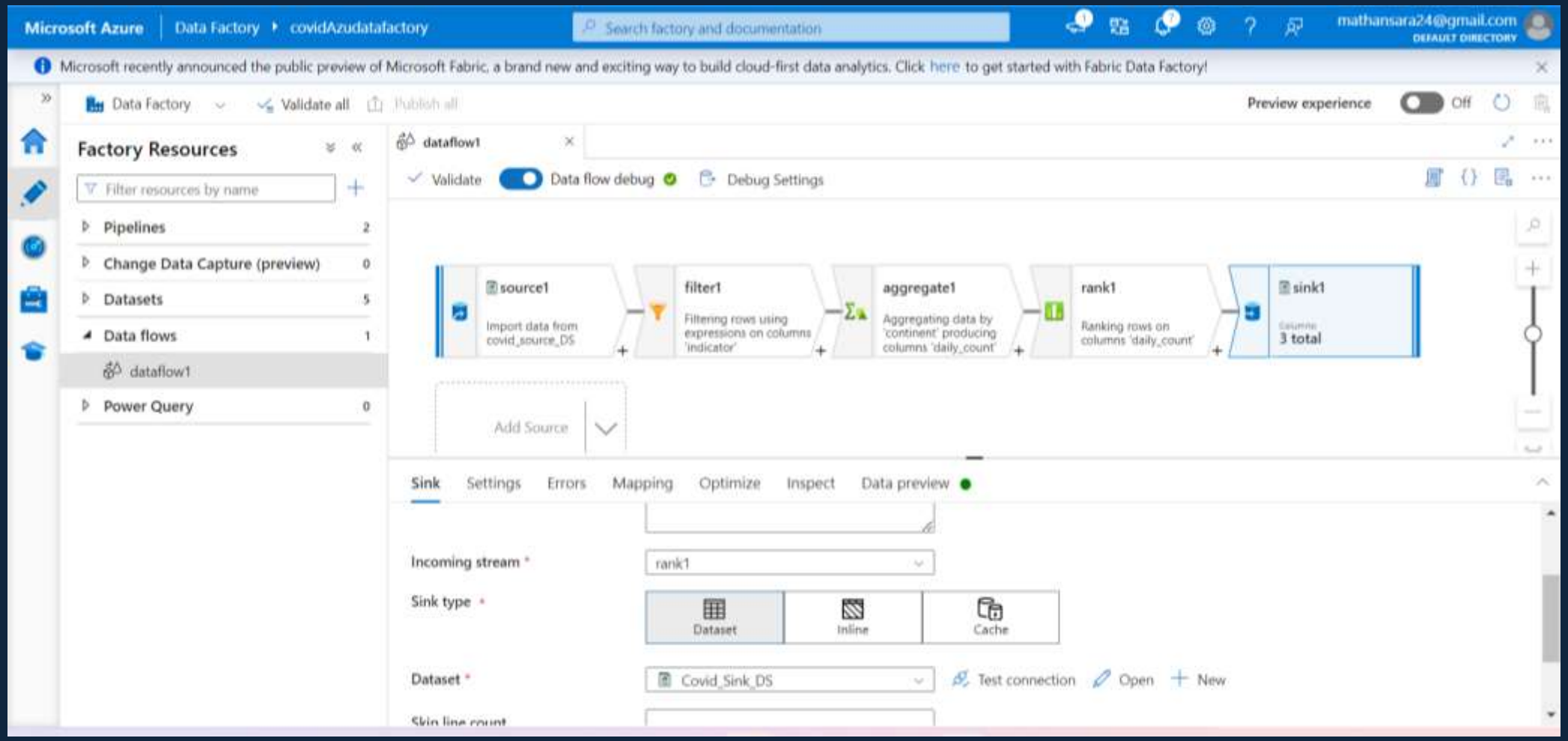
DATA FLOW – AGGREGATE



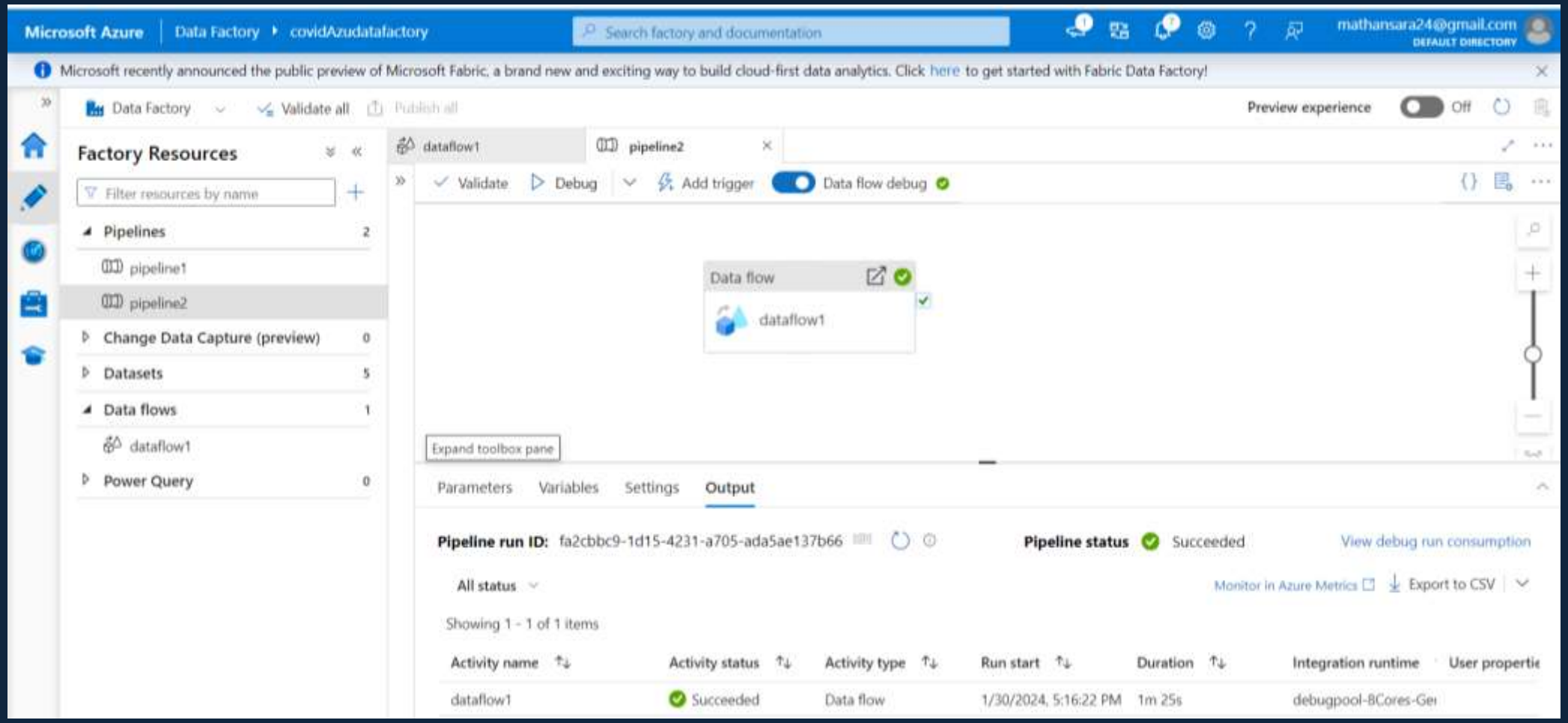
DATA FLOW – RANK



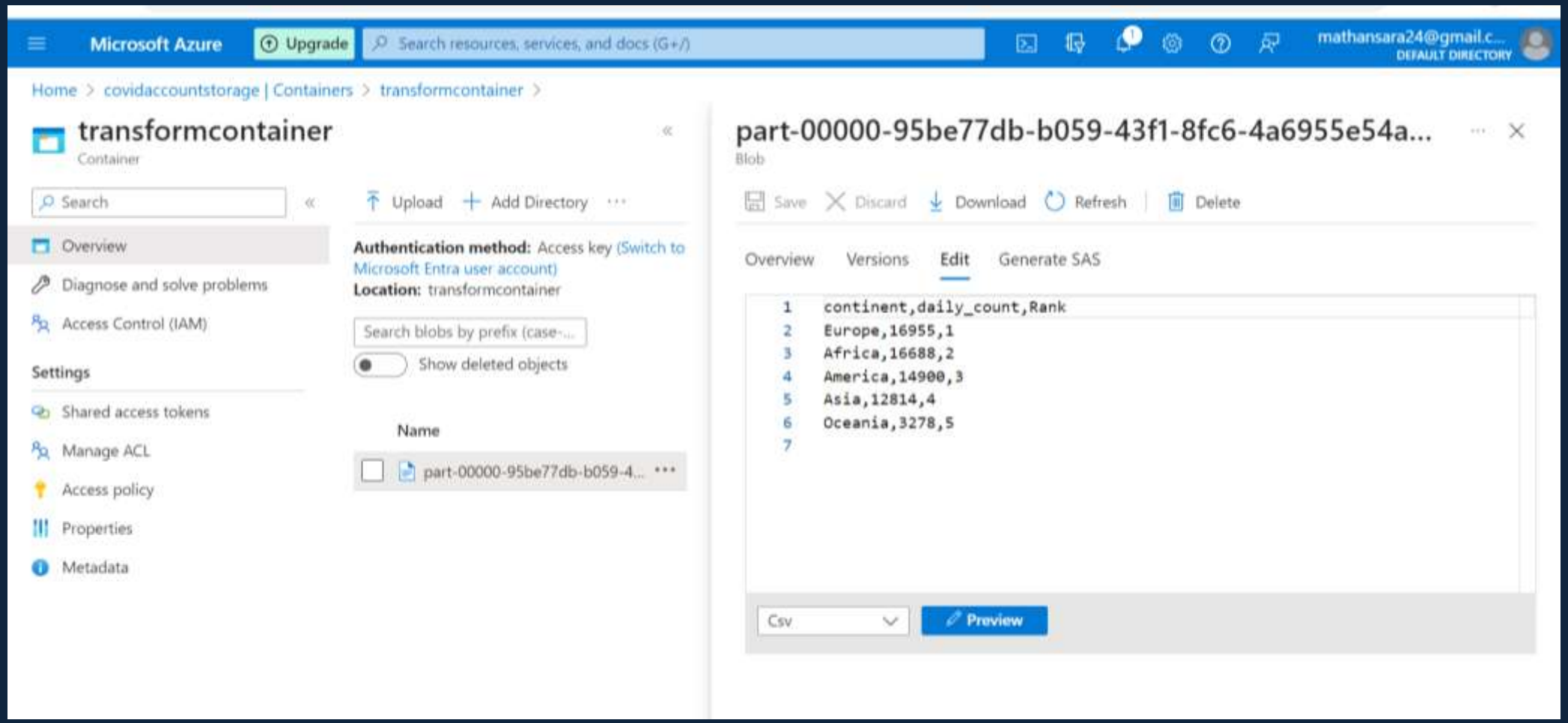
DATA FLOW – SINK



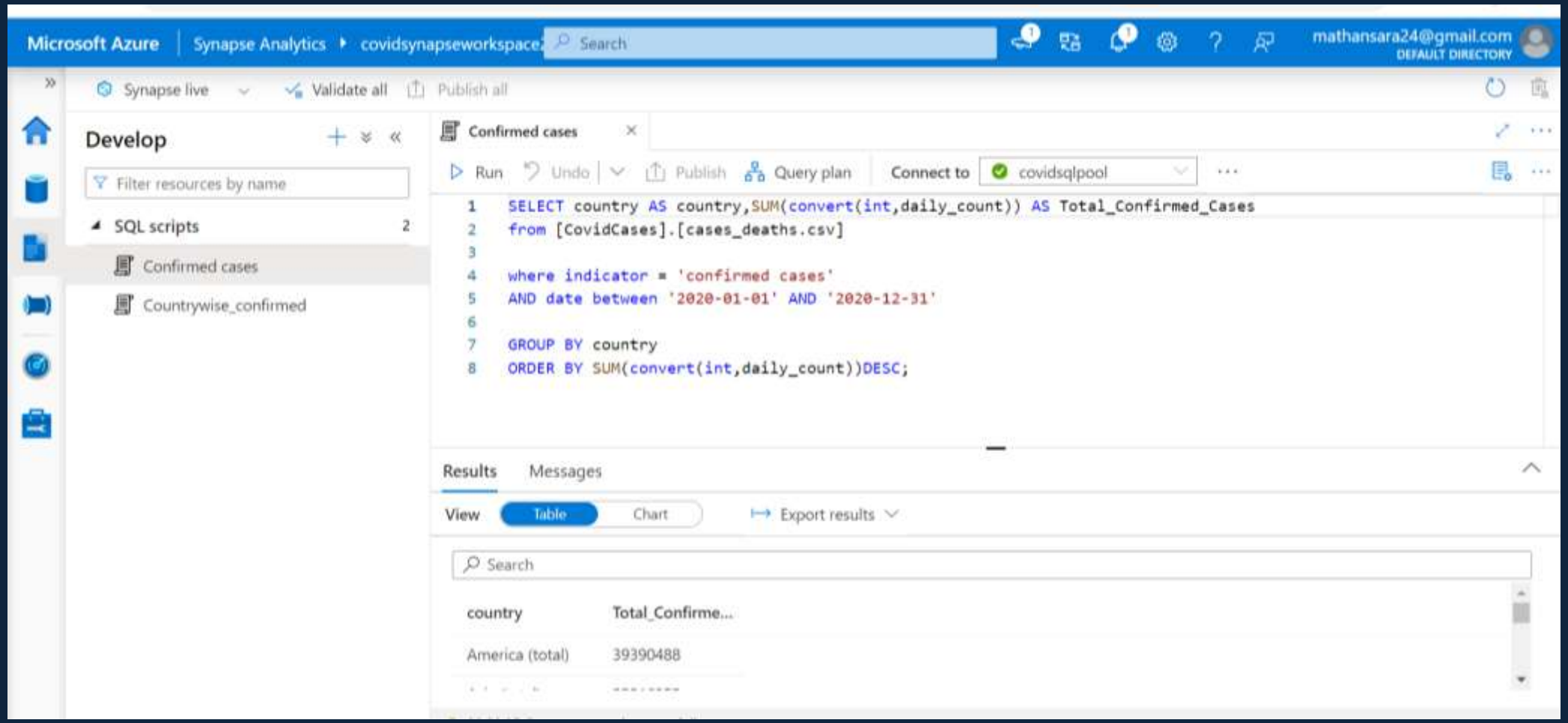
DATA FLOW PIPELINE



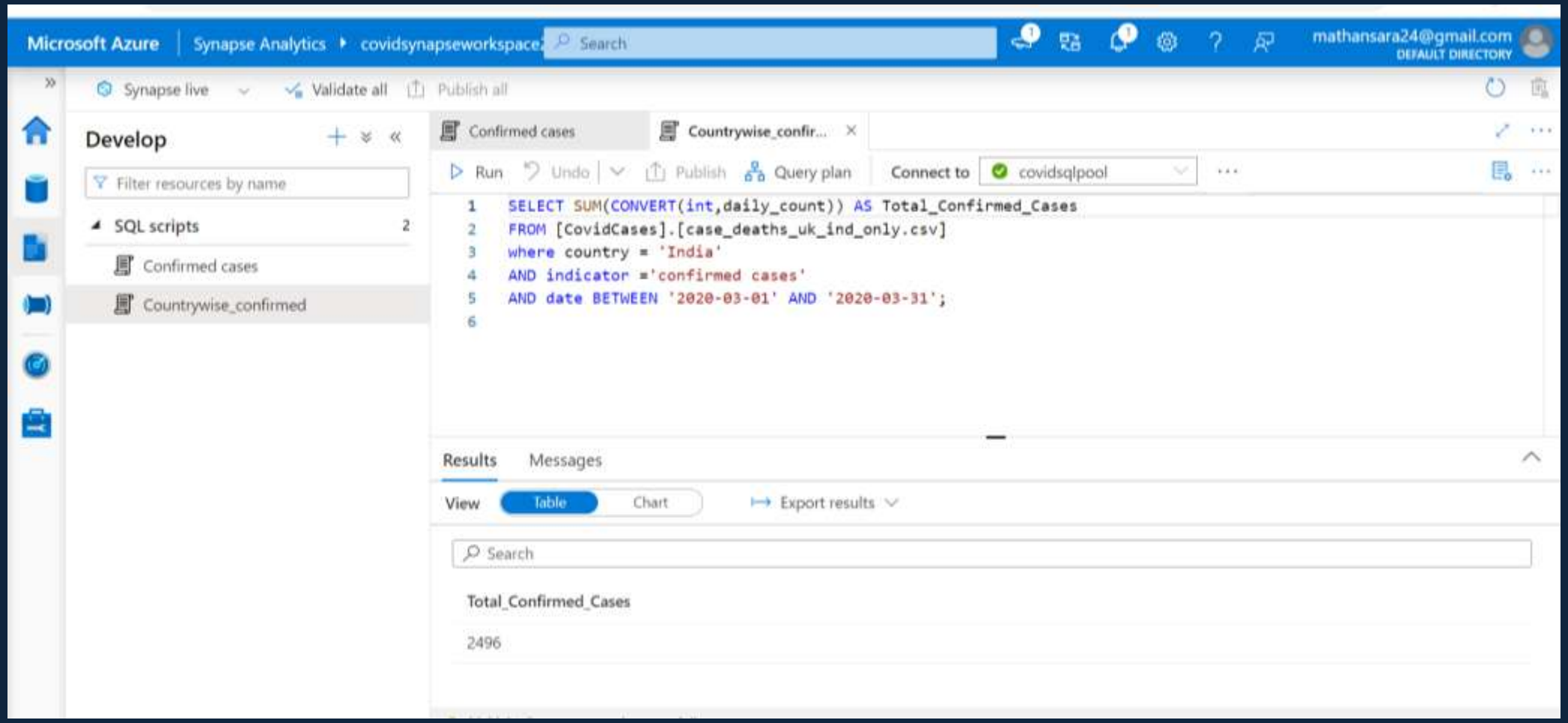
DATAFLOW TRANSFORMATION RESULT



DATA VERIFICATION – QUERYING TO SHOW COUNTRY WISE CONFIRMED CASES IN 2020



QUERYING FOR CONFIRMED CASES OF COVID IN INDIA IN SYNAPSE



SUMMARY

Resources have been successfully created and data has been loaded into Synapse for analysis on Azure.